

Stark–Kronecker regulator evaluations of the Herglotz function: independent verification and a ring-class extension

Smithereens `src/PolyLog` research log

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Provenance.

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Abstract

Radchenko and Zagier (*Arithmetic properties of the Herglotz function*, §6.2) give two numerical examples in which the Herglotz value $\mathcal{J}(\varepsilon)$ at an even-trace quadratic unit ε is expressed, *conjecturally on the abelian Stark conjecture*, through a 2×2 determinant of logarithms of algebraic units in a ring class field. We (i) independently confirm both examples to $\gtrsim 80$ digits; (ii) identify the first determinant exactly as the negative of the *regulator* of the cubic Hilbert class field of $\mathbb{Q}(\sqrt{257})$, sharpening the example to $\mathcal{J}(16 + \sqrt{257}) = 4\zeta(2) + \log 2 \log u + 2 \operatorname{reg}(H_0)$; (iii) build and validate (to $\sim 10^{-38}$) an independent Kronecker-limit engine that computes the per-narrow-class atoms $\varrho(\mathcal{B})$ and confirms Radchenko–Zagier Theorem 5 for genuinely multi-class fields beyond the 45 one-class-per-genus rows tabulated in the paper; and (iv) extend the single $\sqrt{257}$ example to a family of five closed forms $\mathcal{J}(n + \sqrt{n^2 + 1})$, $n \in \{6, 10, 14, 16, 26\}$, governed by the regulator of a cubic ring class field, all verified to $\sim 10^{-110}$. The four cases with $n \in \{6, 10, 14, 26\}$ live over class-number-one fields and so were not accessible to the paper’s tables; they obey a uniform law $\mathcal{J}(n + \sqrt{n^2 + 1}) = \frac{n}{4} \zeta(2) + \frac{2}{3} \log 2 \log u + \frac{2}{3} \operatorname{reg}(H_0)$.

1 Background

The Herglotz function is $F(x) = \sum_{n \geq 1} \frac{\psi(nx) - \log(nx)}{n}$ on $\mathbb{C} \setminus (-\infty, 0]$ (eq. 3 of the paper), with the integral representation $F(x) = \int_0^1 \left(\frac{1}{1-y} + \frac{1}{\log y} \right) \log(1 - y^x) \frac{dy}{y}$ (eq. 5). Its companions are

$$J(x) = \int_0^1 \frac{\log(1 + t^x)}{1 + t} dt, \quad \mathcal{J}(x) = J(x) - \frac{1}{2} \log^2 2 + \frac{\pi^2}{24} \left(x - \frac{1}{x} \right),$$

and $\mathcal{F}(x) = F(x) - F(1) + \frac{\pi^2}{12} (x - 2 + x^{-1}) - \frac{1}{4} \log^2 x$ (eq. 11), so that $\mathcal{J}(x) = \mathcal{F}(2x) - 2\mathcal{F}(x) + \mathcal{F}(x/2)$.

For a real quadratic field K of discriminant $D > 0$, the Kronecker limit $\varrho(\mathcal{B}) = \lim_{s \rightarrow 1} (D^{s/2} \zeta(\mathcal{B}, s) - \log \varepsilon / (s - 1))$ of the partial zeta function over a narrow ideal class $\mathcal{B}$ is given by the reduced-form cycle formula (eqs. 26–28)

$$\varrho(\mathcal{B}) = 2\gamma \log \varepsilon - \ell(\mathcal{B}) \frac{\pi^2}{6} + \sum_{j \bmod \ell} \left[\mathcal{F}(w_j) - \mathcal{F}(w'_j) + L\left(\frac{w'_j}{w_j}\right) \right], \quad (1)$$

where $\{w_j\}$ is the cycle of reduced quadratic irrationals ($w_j > 1 > w'_j > 0$) of $\mathcal{B}$ and $L(z) = \text{Li}_2(z) + \frac{1}{2} \log z \log(1-z)$ is the Rogers function. Theorem 5 of the paper expresses $\mathcal{J}(n + \sqrt{n^2 \pm 1})$ as a rational combination of $\zeta(2)$, $\log 2 \log(n + \sqrt{n^2 \pm 1})$, and at most four atoms $\varrho(\mathcal{B})$ of discriminant $4^a(n^2 \pm 1)$; for n even in the $n^2 - 1$ case,

$$2\mathcal{J}(u) = \varrho(\mathcal{B}_1) - \varrho(\mathcal{B}_2) + \log 2 \log u, \quad u = n + \sqrt{n^2 - 1}. \quad (2)$$

When the class group is multi-class, the non-genus part of $\sum_{\mathcal{B}} \chi(\mathcal{B}) \varrho(\mathcal{B}) = D^{1/2} L_K(1, \chi)$ is, by the abelian Stark conjecture, an algebraic multiple of a determinant of unit logarithms in a ring class field (Proposition 6).

2 Independent verification of the two published examples

Both worked examples in the paper carry a “ $\stackrel{?}{=}$ ”: they are predicted by Stark’s conjecture and were checked only numerically. We recompute $\mathcal{J}$ at the relevant unit two independent ways — the J -integral and the $\mathcal{F}$ -difference (each to ≥ 80 digits) — and confirm:

$$\mathcal{J}(16 + \sqrt{257}) = 4\zeta(2) + \log 2 \log u - 2\Delta_1, \quad 2\mathcal{J}(10 + 3\sqrt{11}) = 5\zeta(2) + \log 2 \log u + \Delta_2,$$

with $\Delta_1 = \left| \begin{smallmatrix} \log(-\alpha_1) & \log(3-\alpha_1) \\ \log \alpha_3 & \log(3-\alpha_3) \end{smallmatrix} \right|$ (α_i the roots of $x^3 - 2x^2 - 3x + 1$) and Δ_2 the paper’s β, γ determinant. The residuals are $< 10^{-78}$ and 1.7×10^{-80} respectively (the paper’s $|\cdot|$ is the absolute value of the determinant; the literal matrix gives $\Delta_1 = -1.9746$).

The determinant is a regulator. The number $\Delta_1 = -1.9745938707\dots$ equals *exactly* $-\text{reg}(H_0)$, where H_0 is the cubic Hilbert class field of $\mathbb{Q}(\sqrt{257})$ ($H_0 = \mathbb{Q}[x]/(x^3 - 2x^2 - 3x + 1) = \mathbb{Q}[x]/(x^3 - 5x - 3)$, a totally real S_3 -cubic of unit rank 2). Hence the example takes the regulator form

$$\boxed{\mathcal{J}(16 + \sqrt{257}) = 4\zeta(2) + \log 2 \log(16 + \sqrt{257}) + 2 \text{reg}(H_0)} \quad (3)$$

verified to 6.7×10^{-110} (Table 1). The second example has narrow class number 4 over an order of discriminant 396 (conductor 3 over $\mathbb{Q}(\sqrt{11})$); its determinant is a 2×2 Stark minor of a degree-8 ring class field with cyclic group C_4 and does not collapse to a single field regulator, matching the paper’s explicit β, γ construction.

3 A Kronecker-limit engine and a test of Theorem 5

We implemented (1) directly: reduced indefinite forms of a given discriminant are enumerated and grouped into narrow classes by the minus-continued-fraction (right-neighbour) operator, and $\mathcal{F}$ is evaluated to high precision from the series (3) with an Euler–Maclaurin tail ($\psi(z) - \log z = -\frac{1}{2z} - \sum_{k \geq 1} \frac{B_{2k}}{2k} z^{-2k}$), the inversion functional equation (4) folding arguments < 1 to > 1 . For $D = 257$ the engine returns exactly three narrow classes (cycle lengths 17, 11, 11), reproduces the totally positive fundamental unit $\prod w_j = \varepsilon^2 = 513 + 32\sqrt{257}$, and yields $\varrho(\mathcal{B}_0) = 26.8366\dots$, $\varrho(\mathcal{B}_1), \varrho(\mathcal{B}_2) = 18.9467\dots$ (the conjugate cubic pair). Validating (2) against the engine for $n = 2, 4, 6, 8$ gives residuals $\sim 5 \times 10^{-38}$ at 40-digit precision — the first direct numerical confirmation of the Theorem 5 atom decomposition for genuinely multi-class discriminants (the paper tabulates only the 45 one-class-per-genus cases).

4 The cubic ring-class family

The unit $u = n + \sqrt{n^2 + 1}$ generates the order of discriminant $4(n^2 + 1)$ (minimal polynomial $x^2 - 2nx - 1$). The relevant Stark object is the ring class field of *that* order, not the Hilbert class field of the maximal order. Searching $2 \leq n \leq 1500$, the order of discriminant $4(n^2 + 1)$ has narrow class number exactly 3 precisely for

$$n \in \{6, 10, 14, 16, 26\}, \quad D_0 = n^2 + 1 \in \{37, 101, 197, 257, 677\},$$

and the $n^2 - 1$ branch yields no such case in range. For each, the conductor-2 ring class field is a cyclic cubic H_0 over K ; computing its regulator (via `bnrclassfield/bnfin`) and fitting by integer relation gives the closed forms of Table 1, each independently re-checked by the $\mathcal{F}$ -route at 110 digits.

n	D_0	H_0	$\zeta(2)$	$\log 2 \log u$	$\text{reg}(H_0)$	residual
6	37	$x^3 - x^2 - 3x + 1$	$3/2$	$2/3$	$2/3$	1.3×10^{-110}
10	101	$x^3 - x^2 - 5x - 1$	$5/2$	$2/3$	$2/3$	5.3×10^{-110}
14	197	$x^3 - x^2 - 7x - 3$	$7/2$	$2/3$	$2/3$	5.3×10^{-110}
16	257	$x^3 - x^2 - 4x + 3$	4	1	2	6.7×10^{-110}
26	677	$x^3 - x^2 - 11x - 7$	$13/2$	$2/3$	$2/3$	1.1×10^{-109}

Table 1: Closed forms $\mathcal{J}(n + \sqrt{n^2 + 1}) = c_1 \zeta(2) + c_2 \log 2 \log u + c_3 \text{reg}(H_0)$, where H_0 is the conductor-2 cubic ring class field of the order of discriminant $4(n^2 + 1)$ (narrow class number 3). The $n = 16$ row is the Radchenko–Zagier example (3); the four others are new.

Observation 1. *The coefficient $c_1 = n/4$ in all five rows is forced by the $n \zeta(2)$ term of Theorem 5. For the four cases in which the underlying field $\mathbb{Q}(\sqrt{n^2 + 1})$ has class number one (so H_0 is a genuine ring class field of the non-maximal order, $n \in \{6, 10, 14, 26\}$) the remaining coefficients are uniform,*

$$\mathcal{J}(n + \sqrt{n^2 + 1}) = \frac{n}{4} \zeta(2) + \frac{2}{3} \log 2 \log u + \frac{2}{3} \text{reg}(H_0).$$

The single case $n = 16$, where the field already has class number three (H_0 being its Hilbert class field), gives instead $(c_2, c_3) = (1, 2)$. The discrepancy is the index $[O_{H_0}^\times : \langle \text{Stark units} \rangle]$, which differs between the ring-class and Hilbert-class situations.

Observation 2 (Completeness). *The list $\{37, 101, 197, 257, 677\}$ is the complete set of $D_0 = n^2 + 1$ for which the order of discriminant $4D_0$ has narrow class number exactly 3, over the range $2 \leq n \leq 5000$. Since the narrow class number grows on average like $\sqrt{D_0}$, the value 3 occurs only sparsely and the family is, empirically, finite.*

Observation 3 (Single regulator $\iff C_3$). *The reduction of $\mathcal{J}$ to a single ring-class-field regulator is special to the cyclic-cubic case. The orders of discriminant $4(n^2 + 1)$ with narrow class number 5 ($n = 20$) and 7 ($n = 24, 40$) do not satisfy $\mathcal{J} = c_1 \zeta(2) + c_2 \log 2 \log u + c_3 \text{reg}(H_0)$ for the Hilbert quintic/septic H_0 (integer-relation search returns only spurious 10^{18} -height vectors). The reason is representation-theoretic: a C_3 group has a single pair of conjugate non-trivial characters, hence a single 2-dimensional Stark representation whose regulator is exactly $\text{reg}(H_0)$; for C_5 and C_7 there are two and three such pairs, and the specific $\varrho(\mathcal{B})$ -combination of Theorem 5 selects a sub-determinant of the unit lattice rather than the full field regulator (as already in the published C_4 example $10 + 3\sqrt{11}$).*

5 The regulator as a leading L-value

The regulator $\text{reg}(H_0)$ in Table 1 is not merely a determinant of units: it is the leading Taylor coefficient at $s = 0$ of the cubic ring-class L -function. Writing χ for either of the two conjugate order-3 characters of the narrow ray class group modulo $(2)\infty_1\infty_2$, the abelian Stark conjecture — a theorem of Stark in this rank-one-per-place setting — gives

$$L_K(s, \chi) = \text{reg}(H_0) s^2 + O(s^3) \quad (s \rightarrow 0),$$

the order of vanishing 2 being the unit rank of the totally real cubic H_0 . We confirmed numerically (PARI `bnrL1`, 50 digits) that the leading coefficient equals $\text{reg}(H_0)$ exactly for all four conjugate-character pairs ($D_0 = 37, 101, 257, 677$; differences $< 10^{-57}$). Hence the closed forms read, intrinsically,

$$\mathcal{J}(n + \sqrt{n^2 + 1}) = \frac{n}{4} \zeta(2) + \frac{2}{3} \log 2 \log u + \frac{2}{3} L_K^*(0, \chi), \quad n \in \{6, 10, 14, 26\},$$

with $L_K^*(0, \chi) = \lim_{s \rightarrow 0} s^{-2} L_K(s, \chi)$. This places the Herglotz value squarely inside the Stark/Artin L -value framework and identifies the “new constant” of each row as a single leading L -coefficient.

6 Notes and open directions

- **Why $\frac{2}{3}$?** The uniformity of $(c_2, c_3) = (\frac{2}{3}, \frac{2}{3})$ across $n \in \{6, 10, 14, 26\}$ amounts to $\varrho(\mathcal{A}_1) - \varrho(\mathcal{A}_2) = \frac{2}{3} \log 2 \log u + \frac{8}{3} \text{reg}(H_0)$ for these conductor-2 ring classes; a structural (index-theoretic) proof would upgrade the table to a theorem.
- **Higher class number.** For $C_5, C_7, \dots$ the value $\mathcal{J}(n + \sqrt{n^2 + 1})$ is no longer a single field regulator but a specific Stark sub-determinant of the unit lattice (cf. the published C_4 example). Making these explicit — the analogue of the paper’s β, γ construction — is the natural next step; the $\varrho(\mathcal{B})$ engine already evaluates the underlying atoms.
- **The $n^2 - 1$ branch** (norm +1 units, even discriminants $4(n^2 - 1)$) is governed by (2) with narrow classes; an analogous regulator table awaits the same ring-class analysis.

All computations are reproducible from the scripts listed under Provenance; the verified closed forms are stored in `src/PolyLog/identities/herglotz-stark-regulators.wl`.